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# An algebraic method for classifying closed and connected surfaces

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**Abstract**

For a given planar diagram of a closed & connected surface, we establish an algebraic method to determine the type of the surface. This method transforms cutting and gluing operations, which are described in the constructive proof of the classification of compact & connected surfaces (see [1] page 79 for example), on the planar diagram of the given surface. The advantage of our approach is that it does not require drawing any diagram, it is less prone to errors, and most importantly it can easily be coded as a computer program.

## 1-) Motivation

One of way of classifying a compact & connected surface is to put a planar diagram on it, and apply the algorithm outline in the classification theorem - see [1] page 79 for example - to determine the type of the surface.However, that algorith requires drawing lots of diagrams, it is prone to errors, and it is slow.However, our method describes a way of using the same algorithm just by manipulating the names of the edges of the given planar diagram as if they are elements in a group.

## 2-) Preliminary Observations

For a given surface and a planar diagram belonging to that surface, w.l.o.g, we may assume that that all the vertices in the diagram are the same; otherwise, we can first eliminate any distinct vertices, as also described in the mentioned proof above, so a typical (simple) planar diagram will loke like this:

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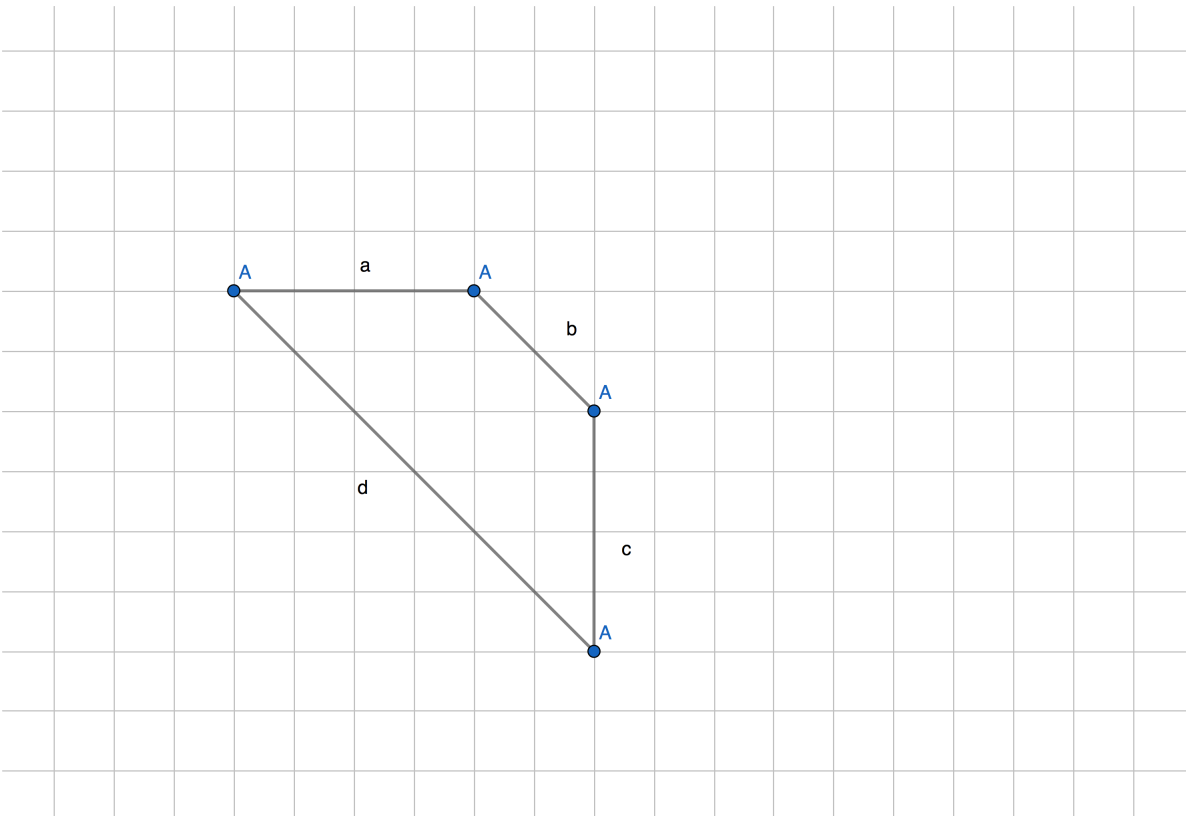


Now, observe that if we were to read the labes of the edges, we could have said, $abcd$, $bcda$, $cdab$, etc. (to be explicit, these are the other possible cases: $dabc$, $d^{-1}c^{-1}b^{-1}a^{-1}$, $c^{-1}b^{-1}a^{-1}d^{-1}$, $b^{-1}a^{-1}d^{-1}c^{-1}$, $a^{-1}d^{-1}c^{-1}b^{-1}$), i.e as long as the order of the edges are the same, there is not preferred specific edge that we need to start reading, nor there is a preferred orientation for how to read the labels of edges.Note that, for example, $a^{-1}$ represents that while we are reading the edge $a$, we going in the reverse direction in the direction of the edge $a$.Therefore, if we were to represent the given surface with an expression containing its edges with their order as one of those given above, $abcd$ and $d^{-1}c^{-1}b^{-1}a^{-1}$ would yield the same surface as long as we also specify with respect to which orientation that we are reading the names of the edges.

Now consider the surface given in Figure 1.1,

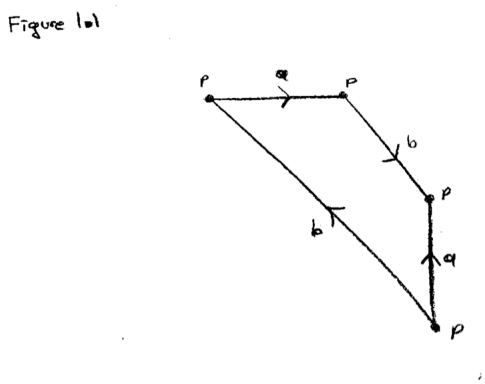


Figure 1.1

and let first fix the orientation which we will read our labels as the counterclockwise direction.In particular, one of the possible way to read this surface would be $aba^{-1}b^{-1}$.Then, if we were to cut a given surface, a new edge $e$ would be created as it is shown in figure 1.2.

Figure 1.2

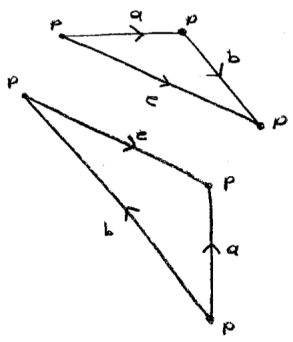


Figure 1.2

Now, if we read the labels of these two components of the diagrams in the same direction that we have read our orijinal diagram, we can read them as $abe^{-1}$ and $ea^{-1}b$.

Observe that the expressions $abe^{-1}$ and $ea^{-1}b$ are in such a form such that while we are cutting, we created $ee^{-1}$ pair, and gave one piece to each component.Conversely, if we were to glue the pieces along $e$, geometrically, each component has to have an edge $e$, and they should be in the same direction (geometrically), but this means, in our language, we should have $e$ in one component, and $e^{-1}$ in the other component. Note tha, since it does not matter from which edge we start reading our labels, we can represent the components as in the form $...e$ and $e^{-1}$.....The reason why we need such a representation will be explained.

Furthermore, if we reflect one of the pieces along one of the edges, as it is shown in the figure 1.3, the representation of this piece changes from $abe^{-1}$ to $eb^{-1}a^{-1}$; therefore given such a representation of diagram, reversing the order of edges while replacing each edge name with its inverse, i.e $ab \to b^{-1}a^{-1}$, corresponds to reflection diagram along one of its edges - along which it is done does not change the result.

As for anyone who is familiar with Algebra, these observation are similar to the properties of algebraic structures on sets, so we are going to try to model these operations in a similar way.

Figure 1.3

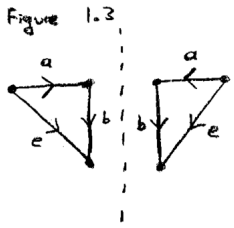


Figure 1.3

### 3-) Main definitions and Rules

For a given planar diagram of a surface, let fix an orientation - clockwise or counterclockwise -, and read the labels of the edges of the surface with that orientation.

i-) Then cutting the surface along a new edge, say $x$, will introduce a new edge, say $e$, in the representation of the components such that one of component will get $e$, and the other will get $e^{-1}$ in their representation.Geometrically speaking this operation directly captures the cutting operation completely.

ii-) To glue to pieces along an edge, say along $x$, we need to first represent the components in such a way that $x$ and $x^{-1}$ are in the one of the end of in the representations of the components, such as $...abcx$ and $x^{-1}efg...$, so that after the gluing, we will obtain a suface whose representation is $...abcefg....$

iii-) Any diagram can be reflected along one of its edges - which does not change the diagram geometrically and it is also used in the actualy proof -. The corresponding operation in the representation is to change the order of each edge while replacing each edge with its inverse, i.e $abc \to c^{-1}b^{-1}a^{-1}$ - algebraically it might be interpreted as taking the inverse of the whole expression corresponds to reflecting the diagram along one of its edges, i.e $(abc)^{-1} := c^{-1}b^{-1}a^{-1}$.

It should be noted that this method is just transforms the geometric operations, which are described in the constructive proof of the classification theorem, to algebraic operation. Therefore, in order to use this method effectively, one has first know how to classify the surface geometrically. In other words, we just transform the geometric problem to an algebraic one.

## 1 An Application

Consider the general planar diagram of $\mathbb{K}^2\#\mathbb{T}^2$, and represent it as $cdc^{-1}d^{-1}ba^{-1}ba$.

With the rules that we described above, we can argue that as follows:

by putting the $ba$ in the end to the beginning of representation,

$$cdc^{-1}d^{-1}ba^{-1}ba = bacdc^{-1}d^{-1}ba^{-1},$$

and cutting the surface from the vertex that is in the end of $d$ to $b$ in the beginning.

$$\Rightarrow bacdk \quad \textbf{and} \quad k^{-1}c^{-1}d^{-1}ba^{-1} \Rightarrow k^{-1}d^{-1}c^{-1}a^{-1}b^{-1} \quad \textbf{and} \quad ba^{-1}k^{-1}c^{-1}d^{-1}.$$

Lets glue them along $b$,

$$\Rightarrow k^{-1}d^{-1}c^{-1}a^{-1}b^{-1}ba^{-1}k^{-1}c^{-1}d^{-1} = k^{-1}d^{-1}c^{-1}a^{-1}a^{-1}k^{-1}c^{-1}d^{-1}.$$

Name $a^{-1}a^{-1}$ pair as $[P_1]$, a projective plane component, so we have

$$k^{-1}c^{-1}d^{-1}k^{-1}d^{-1}c^{-1}[P_1],$$

cutting the surface from the end of the leftmost $c^-1$ to $k^{-1}$ in the beginning,

$$\Rightarrow k^{-1}c^{-1}e^{-1} \quad \textbf{and} \quad ed^{-1}k^{-1}d^{-1}c^{-1}(P_1),$$

then reflecting the first component, and gluing them along $c$,

$$\Rightarrow kec \quad \textbf{and} \quad c^{-1}[P_1]ed^{-1}k^{-1}d^{-1} \Rightarrow [P_1]ed^{-1}k^{-1}d^{-1}ke.$$

Then cutting from the end of the leftmost $e$ to the beginning of the projective plane component,

$$\Rightarrow [P_1]er \quad \textbf{and} \quad r^{-1}d^{-1}k^{-1}d^{-1}ke$$

then gluing them along $e$

$$\Rightarrow r[p_1]e \quad \textbf{and} \quad e^{-1}k^{-1}dkdr \rightarrow r[P_1]k^{-1}dkdr,$$

Name $rr$ component as $[P_2]$, another projective plane component, so we have

$$[P_2][P_1]k^{-1}dkd.$$

Now, by curring surface from the end of left most $d$ to the begining of the second projective plane component,

$$[P_2][P_1]k^{-1}dkd = [P_2][P_1]k^{-1}dr \quad \textbf{and} \quad r^{-1}kd$$

then gluing them along $d$, we get

$$\Rightarrow r[P_2][P_1]k^{-1}d \quad \textbf{and} \quad d^{-1}k^{-1}r \rightarrow [P_2][P_1]k^{-1}k^{-1}rr.$$

Now, name $k^{-1}k^{-1}$ as $[P_3]$ and $rr$ as $[P_4]$, hence we have

$$[P_1][P_2][P_3][P_4],$$

a connected sum of 4 projective plane!

Another application can be found in [2].

## 2 Conlusion:

As it is demonstrated, same algorithm described in the classification proof, the method outline above reduces the problem to an "algebraic" one in a way that it is less time consuming, programmable, and prone less errors.

As we have given in the section 3, each operation that is used in the proof is in is mapped to an algebraic operation on the representation of the surface.Therefore, anything that can be done with those geometric operations can also be done with this method, and maybe more.

# References


[1] Kinsey, C., Topology of Surfaces, 2nd ed.

[2] http://users.metu.edu.tr/e214070/application.png




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